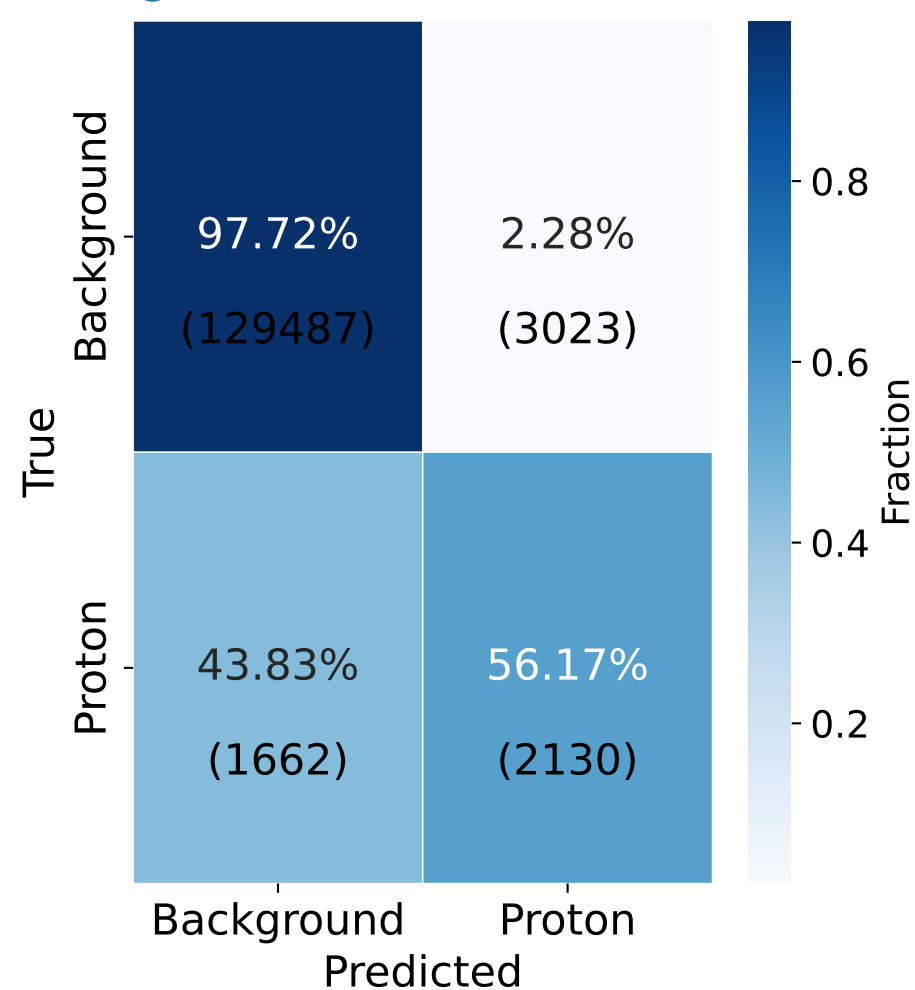


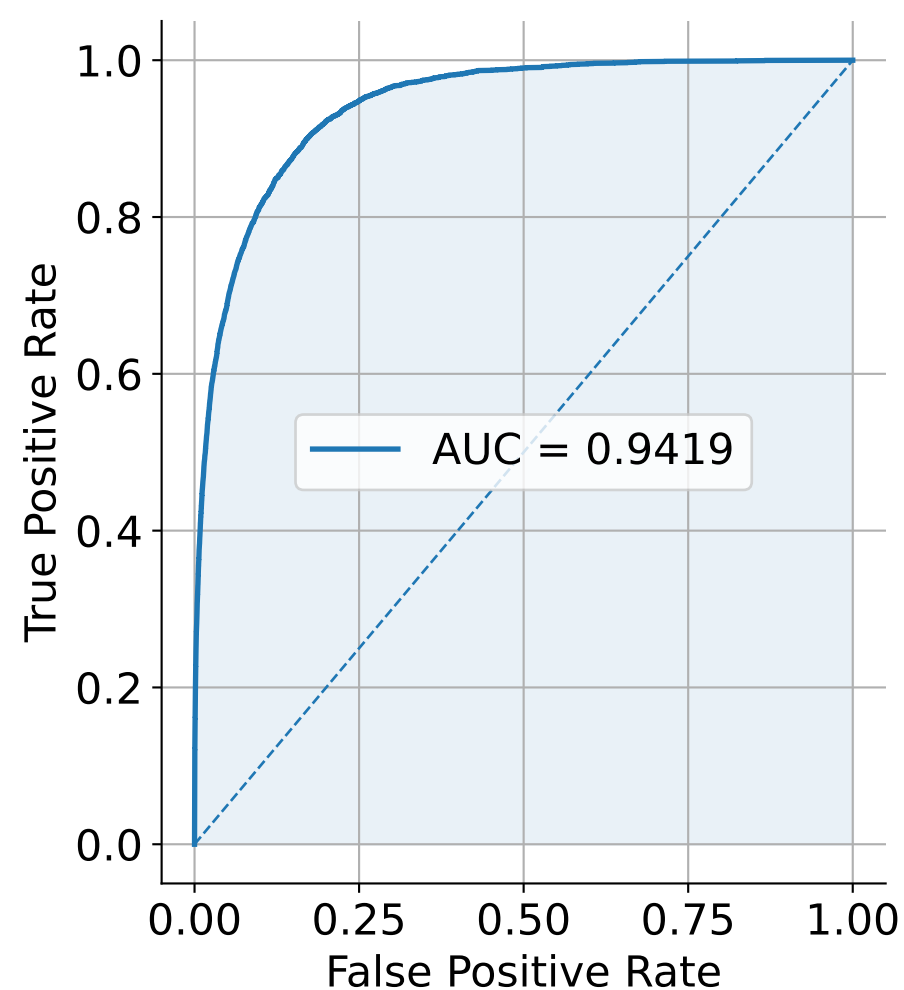
Proton Identification — XGBoost Analysis

[Train S/B=1:35 | Test S/B=1:35 (realistic, all files)]

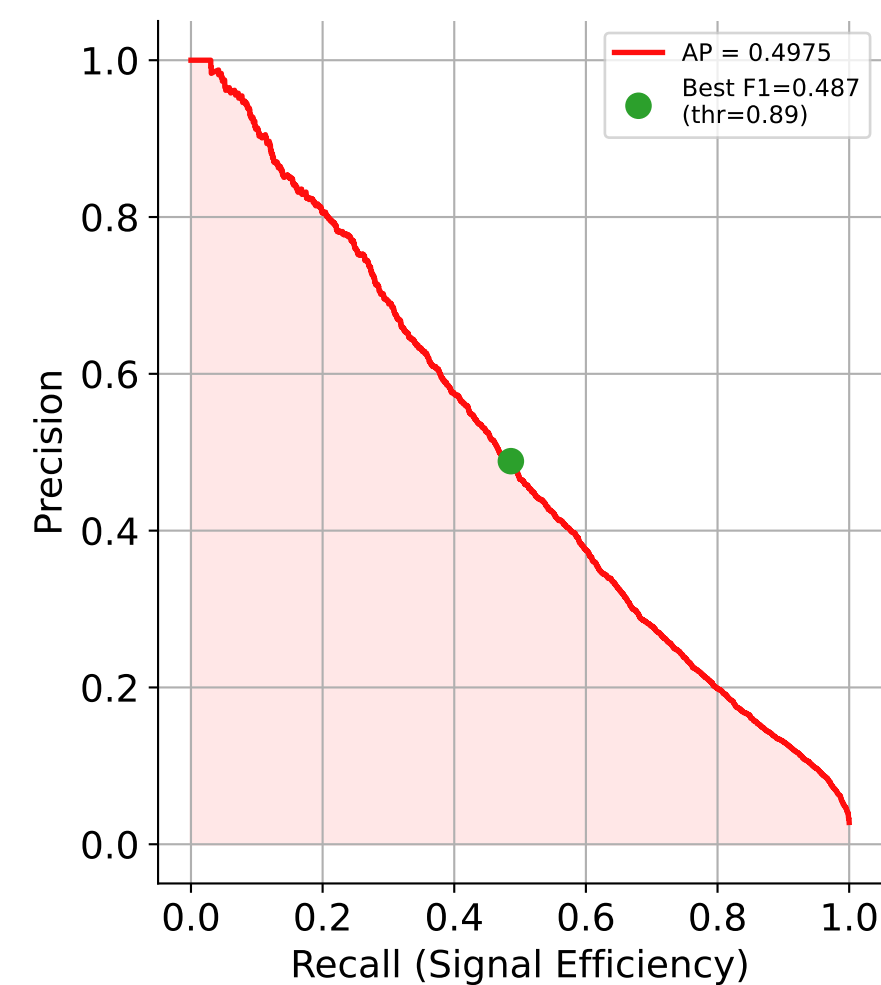
Confusion Matrix
@ Max F1-Score (thr=0.86)



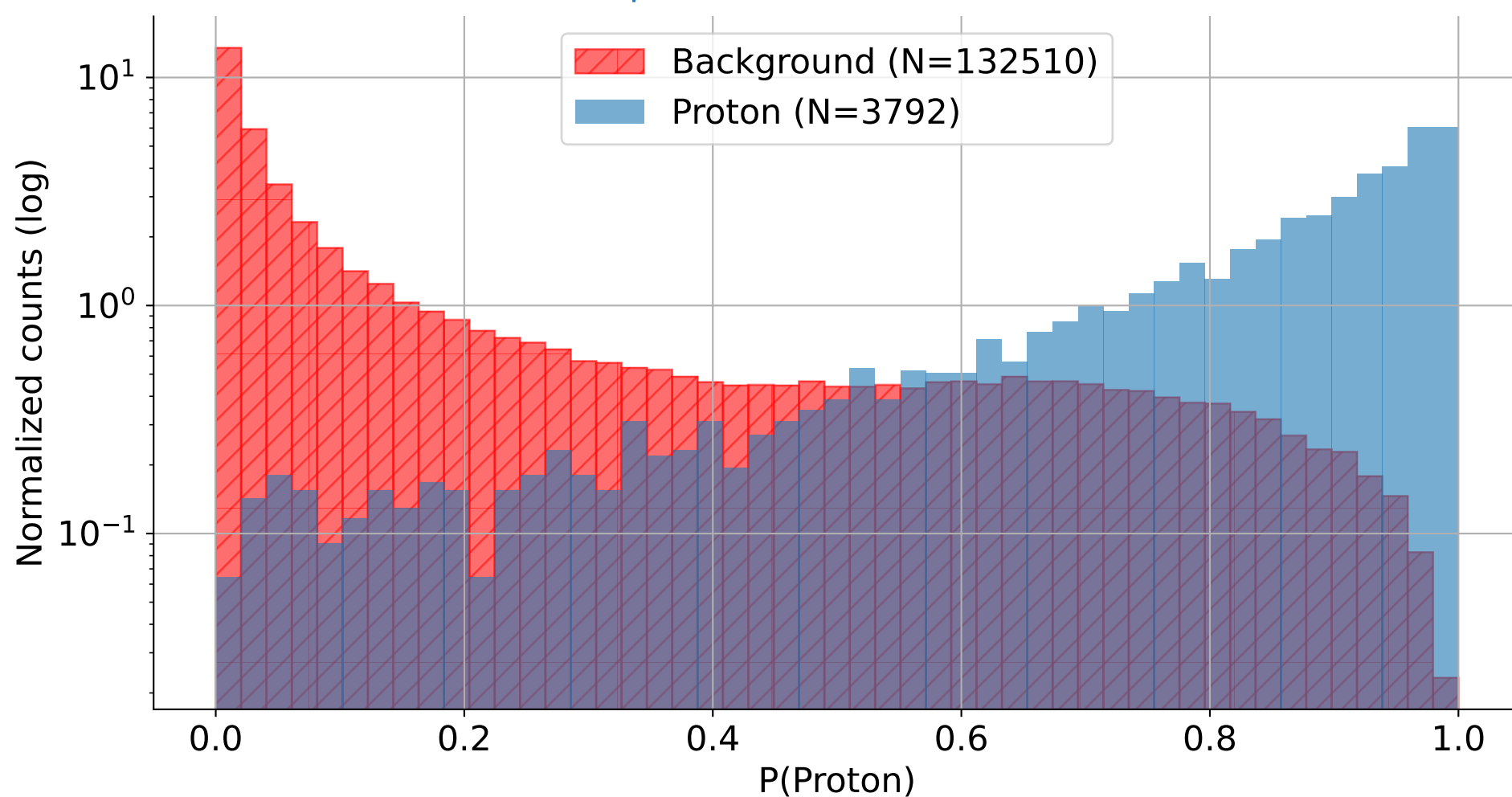
ROC Curve



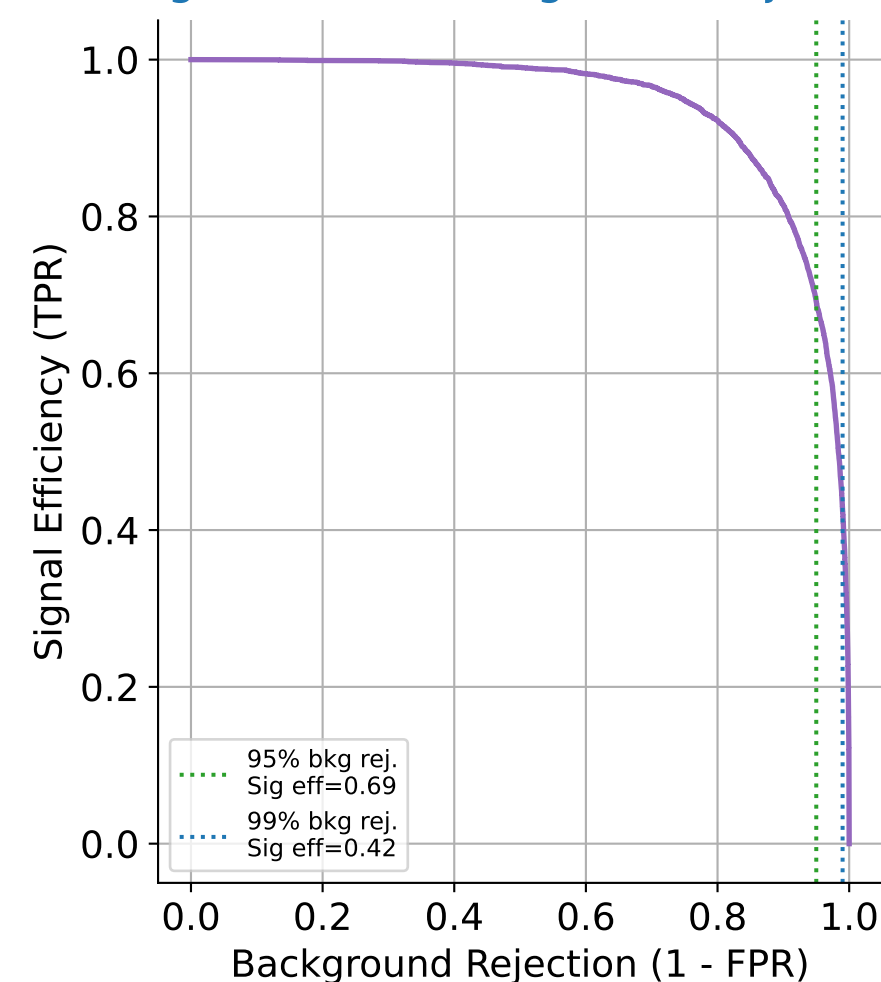
Precision-Recall Curve



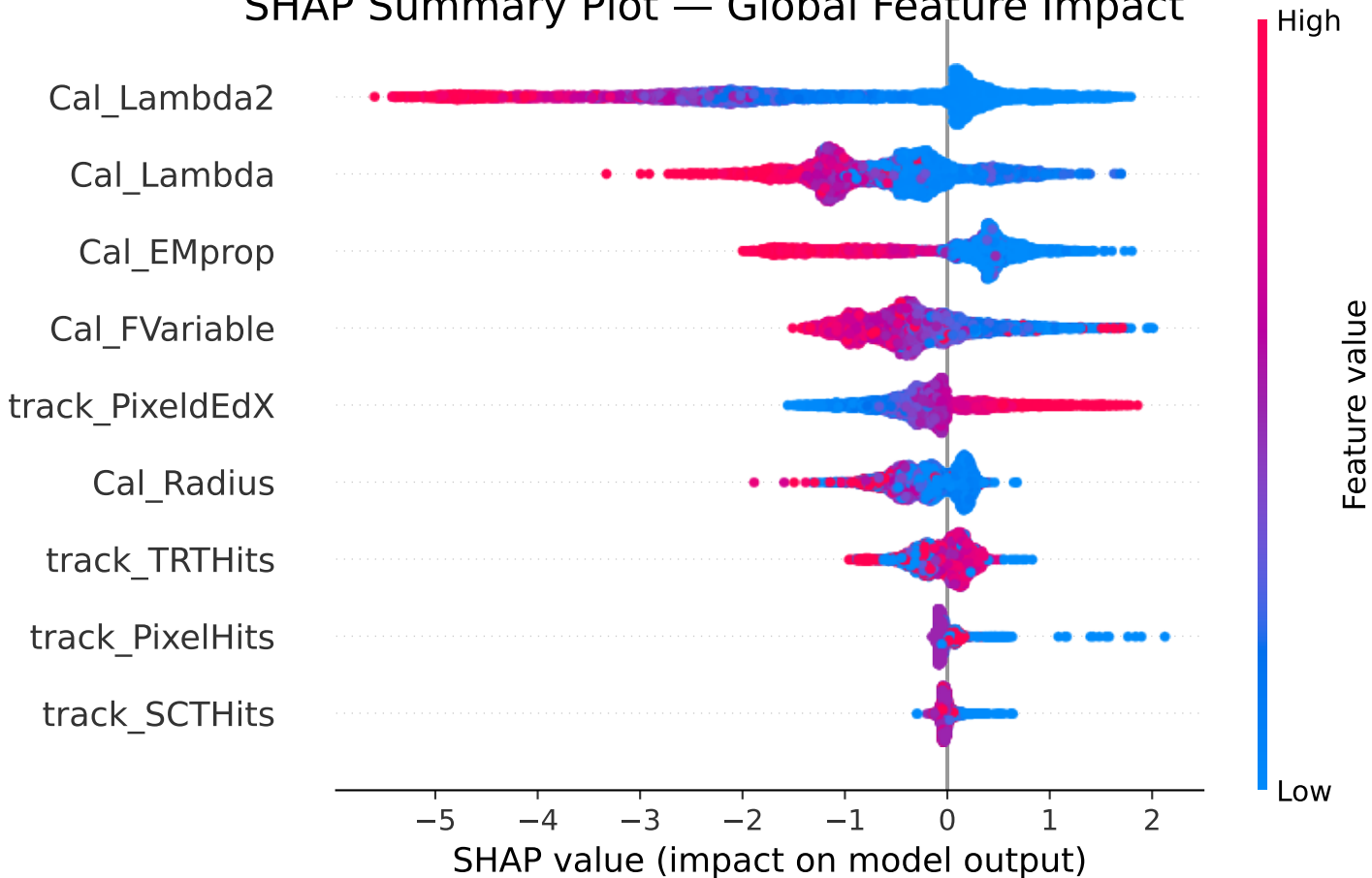
Classifier Response Distribution [Test: realistic S/B]



Signal Eff. vs Background Rejection



SHAP Summary Plot — Global Feature Impact



SHAP Feature Importance (mean |SHAP|)

